

ZIYUE YU

Contact ◇ ziy79@pitt.edu ◇ 412-909-9684

Address ◇ 3700 O'Hara St, Pittsburgh, PA 15213

Google Scholar ◇ ORCID: 0009-0005-7468-0402 ◇ GitHub ◇ Website

EDUCATION

University of Pittsburgh

September 2024 – Present

Ph.D. in Civil and Environmental Engineering

Advisor: Prof. Lei Fang

Southern University of Science and Technology

September 2020 – June 2024

B.Eng. in Mechanical and Aerospace Engineering

Advisor: Prof. Hongyan Yuan

RESEARCH INTERESTS

My research addresses turbulent and coherent transport in weak-inertia flows, where classical inertia-dominated turbulence theory does not apply. Current work spans three directions: scale-to-scale energy transfer at Reynolds numbers of order one; Lagrangian coherent structures in active suspensions; and energy-efficient destabilization of tornado-like vortices, with implications for severe-weather mitigation and infrastructure resilience.

PEER-REVIEWED JOURNAL PUBLICATIONS

1. **Z. Yu**, X. Si, and L. Fang, “Tensor-Geometry-Activated Spectral Energy Transfer at Reynolds Numbers of Order Unity in Two-Dimensional Flows,” *Journal of Fluid Mechanics*, accepted 26 August 2026, in press. Preprint circulated as arXiv:2511.05800 [physics.flu-dyn].

Shows that turbulence-grade scale-to-scale energy transfer can be actively induced at $Re = O(1)$ by enforcing geometric alignment between the turbulent stress and large-scale rate-of-strain tensors, reaching flux magnitudes reported for the weakly turbulent regime without inertial dominance.

Additional peer-reviewed work (undergraduate research, other fields):

2. Wenkai Yu, Jingyi Liu, Xin Li, **Ziyue Yu**, and Hongyan Yuan, “A Novel Hybrid Variable Stiffness Mechanism: Synergistic Integration of Layer Jamming and Shape Memory Polymer,” *IEEE Robotics and Automation Letters*, vol. 9, no. 3, pp. 2734–2741, 2024. (19 citations)
3. Sorour Sadeghzade, Jinrui Cao, Rui Yang, Yuanlong Li, Yanping Li, Dingcong Zhang, Jingyi Liu, **Ziyue Yu**, Liang Fang, and Hongyan Yuan, “Highly Stretchable Alginate/Methylcellulose Hydrogels for 3D Bio-Printing: Photopolymerization Approach Enhancing Structural Integrity,” *Giant*, 2024. (18 citations)

Z. Yu and L. Fang, “Filter-Space Decomposition of the Enstrophy Flux in Two-Dimensional Turbulence: Leonard Artifacts and Cross-Term Transfer.”

First analytical theory of the Leonard artifact in enstrophy flux, showing it vanishes identically in any single-shell flow and requires cross-shell mixing, and identifying the cross term C_2 as the carrier of genuine forward enstrophy cascade. Validated against 1024^2 direct numerical simulation and eight experimental datasets spanning $Re = 11$ – 959 .

Z. Yu^{*}, M. Wu^{*}, and L. Fang, “Two-Way Coupling in Active Suspensions Suppresses the Mobility Selectivity of Particle Accumulation and Induces Non-Monotonic Flow Stabilization.”

Experimental study of the two-way coupling between centimeter-scale active swimmers (*Artemia salina*) and an electromagnetically driven quasi-two-dimensional cellular flow, spanning 85 operating conditions. Establishes a two-way-coupling stability phase diagram in which collective activity can both erode and reinforce the transport skeleton of the flow. ^{*}These authors contributed equally.

Z. Yu and L. Fang, “How Many Steps Does It Take to Kill the Tornado.”

Develops and evaluates an energy-efficient strategy for destabilizing tornado-like vortices by introducing small, directional perturbations that trigger forward spectral energy flux and progressively collapse the vortex core, using three-dimensional velocity fields from CM1 large-eddy simulations.

RESEARCH IN PROGRESS

Cross-Scale Energy and Enstrophy Cascade in Tropical Cyclones

2026 – Present

Applies coarse-grained flux and tensor-geometry diagnostics to ERA5 reanalysis of two Category-5 typhoons (Haiyan 2013, Bavi 2026) across their full life cycles, over scales of 100–1200 km at three pressure levels. Identifies a scale gateway near 500 km through which the storm draws energy from the background field and returns it aloft, and shows that enstrophy cascade is gated by tensor alignment rather than amplitude. Relevant to tropical-cyclone intensity forecasting and coastal infrastructure risk.

Alignment-Selected Inverse Energy Transfer in Low-Reynolds-Number Source Arrays 2026 – Present

Combines OpenFOAM simulations of $N = 1$ – 30 momentum-source arrays with a dimensionless framework to explain how collective alignment, rather than single-source far-field response, produces inverse energy transfer toward large scales.

Dynamic Amplification of Perturbation-Induced Subfilter Stress by Background Flow 2026 – Present

Measures a 7 – $70\times$ dynamic amplification of rod-induced subfilter stress by the background flow, an effect absent under kinematic linear superposition, and maps its dependence on background strain rate.

CONFERENCE PRESENTATIONS

Z. Yu and L. Fang, “Low Reynolds Number Turbulence: Tensor Geometry Drives Turbulence in the Diminishing of Inertia,” 78th Annual Meeting of the APS Division of Fluid Dynamics, Houston, Texas, USA. Interactive Session. *November 2025*

RESEARCH METHODS

Experimental Methods: 2D and 3D Particle Tracking Velocimetry (PTV); Electromagnetically Driven Quasi-Two-Dimensional Flow Facilities; Arduino-Based Circuit Design and Motor Control; Active-Matter Cultivation and Maintenance.

Computational Methods: MATLAB (Advanced); Python (NumPy, SciPy, xarray, PyTorch); R and Java. Coarse-Graining and Subgrid-Scale Flux Analysis of Direct Numerical Simulation (1024^2), Large-Eddy Simulation (CM1), and OpenFOAM Data; FTLE/LCS and Lagrangian Diagnostics; ERA5 Reanalysis Processing (NetCDF, Multi-Level, Storm-Following Grids); Automated End-to-End Analysis Pipelines; Parallel Computing; Git and L^AT_EX.

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant, University of Pittsburgh *2024–2025*
CEE 1402: Fluid Mechanics
CEE 1610: Engineering and Sustainable Development

Research Mentor, pre-college student researcher *2025–2026*
Supervised the experimental and analysis work of a high-school student on active suspensions, resulting in shared first authorship on a manuscript in preparation.

AWARDS AND HONORS

Teaching Assistant of the Year, Department of Civil and Environmental Engineering, University of Pittsburgh *2024–2025*